

SEMESTER-III

UMA034 : OPTIMIZATION METHODS				
	L	T	P	Cr
	3	0	2	4.0
Course Objective: The course aims at integrating the mathematical backgrounds for provide the best solution in linear programming problems and network problems or solutions in deterministic situation. In addition to this, the course introduces decision analysis, queuing theory and Monte Carlo simulation techniques for optimization problems.				
Syllabus Scope of Operations Research: Introduction to linear and non-linear programming formulation of different models. Linear Programming: Geometry of linear programming, Graphical method, Linear programming (LP) in standard form, Solution of LP by simplex method, Exceptional cases in LP, Duality theory, Dual simplex method, Sensitivity Analysis Integer Programming: Branch and Bound Technique Transportation and Assignment Problem: Initial basic feasible solution of balanced and unbalanced transportation problems, optimal solutions, Assignment Problem Goal programming: Introduction to Goal programming, Standard form of linear Goal programming problem, Solution of linear Goal Programming problem by graphical method. Network Models: Construction of Networks, network computations, free floats, critical path method, optimal scheduling Decision Analysis: Decision analysis with minimax (maximin) criteria, Dominance property, Two person zero-sum game, Game with mixed strategies, Graphical method Queuing Theory: Pure birth and death model, Classification of Queuing models, M/M/1 model, Introduction to cost models in queuing Monte Carlo simulations: Mid-square algorithm for random number generation, Introduction to simulation of queuing system				
Laboratory Work Lab experiments will be set in consonance with materials covered in the theory using MATLAB.				
Course Learning Outcomes (CLOs) The students will be able to: 1) formulate and solve linear programming problems using Simplex method and its variants. 2) solve linear goal programming problem graphically. 3) construct and optimize various network models 4) to study two-person zero sum game and its solutions. 5) classify and modeling of queuing system.				
Text Books 1. Operations Research, Swarup, K., Gupta, P. K., Mammohan, Sultan Chand & Sons, 2010. 2. Numerical Optimization and Applications, Chandra, S., Jayadeva, Mehra, A., Narosa Publishing House, 2013. 3. Operations Research-An Introduction, Taha H.A., PHI, 2007.				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

Reference Books

1. Introduction to optimization: Operations Research, Pant J. C., Jain Brothers, 2004
2. Linear Programming and Network flows, Bazaarra Mokhtar S., Jarvis John J. and ShiraliHanif D., John Wiley and Sons, 1990
3. Introductory Operations research, H.S. Kasana and K.D. Kumar, Springer publication, 2004
4. Operations Research- Principles and Practice, Ravindran, D. T. Phillips and James J. Solberg, John Wiley & Sons, Second edn., 2005.

Evaluation Scheme:

Sr.No.	Evaluation Elements	Weightage (%)
1.	MST	30
2.	EST	45
3.	Sessionals (May include assignments/quizzes/projects)	25